

B 1.1.1 Problem specification



B 1.1.1 Construct a problem specification.

A problem specification is the formal definition of what a system must do before any design or coding begins. A well-constructed specification includes a problem statement, constraints and limitations, objectives and goals, input and output specifications, and evaluation criteria. Getting this right determines whether a solution actually solves the right problem.

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Key Questions

What are the five elements of a problem specification, and what does each one define?

What is the difference between objectives and goals, and evaluation criteria - and why does that distinction matter?

How does a problem specification differ from a solution description or an implementation plan?

IB command terms

Construct

Display information in a diagrammatic or logical form. For this topic, that means producing a complete, correctly structured specification for a given problem, not just naming the elements.

What to include

For each element: what it defines, why it is needed, and a concrete example. You must be able to write each element for an unseen scenario.

Introduction

WHY THIS MATTERS - A REAL EXAMPLE

The Mars Climate Orbiter, 1999



NASA lost a \$125 million spacecraft because two engineering teams used different units of measurement: one used pound-seconds, the other newton-seconds, in the same navigation software. Neither team had formally specified what unit the data would be in. The spacecraft entered the Martian atmosphere at the wrong angle and was destroyed.

The failure was not caused by complex engineering. It was caused by an incomplete problem specification, a missing constraint that everyone assumed someone else had defined.

What is a problem specification?

A problem specification is a formal document that defines exactly what a system must do. It describes the problem and the requirements for a solution, but it does not describe how the solution will be built or programmed.

This distinction is important. A specification answers: what problem are we solving, and how will we know when it is solved? Implementation answers: how do we build it? Confusing the two is one of the most common errors students make when writing their IA.

According to the IB Computer Science syllabus, a complete specification includes five elements. Each is examined below.

Key Elements of a Problem Specification

1 Problem statement

A clear description of the problem to be solved

The problem statement defines the problem and explains why it needs to be solved. It describes the current situation, who is affected, and the consequences of the problem. It does not describe the solution or name any technology.

A useful test: if your problem statement could be the first sentence of a solution description, it is written incorrectly.

Weak vs strong

Weak	Strong
"Create a Python program with a database to store student attendance records."	"Teachers at the school currently record attendance on paper. Records are frequently lost, and late reports take several hours to compile manually."

The weak example describes a solution, names a language, and assumes a database - none of which belong in a problem statement. **The strong example** describes the situation, the people affected, and the consequences.

2 Constraints and limitations

The boundaries within which the solution must operate

Constraints are the fixed conditions that the solution cannot change; they restrict what the developer can do. They may be technical, operational, or legal in nature.

CATEGORIES OF CONSTRAINT

- **Technical:** must run on Windows 11; must use less than 500MB of storage; must operate on the school's existing hardware.
- **Operational:** must be completed by the end of the academic year; the developer has no budget for paid libraries or services.
- **Legal/social:** must comply with data protection law; must not store personally identifiable data on external servers.

Note: constraints limit the solution space; they are not goals. "The system must be easy to use" is a goal, not a constraint. "The system must run on devices with no internet

connection" is a constraint.

3 Objectives and goals

The broad aims of the solution

Objectives and goals describe the solution's general purpose and intended outcomes. They state what the solution should achieve in broad terms, without specifying how success will be measured. They set the project's direction.

Objective (broad aim)

"To provide teachers with a faster and more reliable method of recording and reporting student attendance."

Evaluation criterion (same idea, made measurable)

"The system must generate a full class attendance report in under 10 seconds when tested with a class of 30 students."

The key distinction: objectives tell you *what* you are trying to achieve. Evaluation criteria tell you *how you will know* whether you have achieved it. Both are required.

4 Input and output specifications

What data enters the system, and what the system produces

Input specifications define the data that the system will receive, including its type, format, valid range, and source. Output specifications define what the system must produce, including the format and destination of the results. Together, they define the boundary between the system and its surroundings.

EXAMPLE - ATTENDANCE SYSTEM

- **Input:** Student ID (integer, 4 digits, entered via keyboard); attendance status (boolean, present/absent); date (in DD/MM/YYYY format, selected from a calendar).
- **Output:** Attendance report (PDF or on-screen table) listing each student, their attendance status, and percentage attendance for the selected period.

Be precise. "The user enters their name" is insufficient. "The user enters a first name (string, max 50 characters) and a family name (string, max 50 characters)" is a specification.

5 Evaluation criteria

Specific, measurable standards for judging success

Evaluation criteria are the precise, testable standards against which the completed solution will be judged. Each criterion must be specific enough to be tested and yield a clear pass or fail result. In the IB IA, these criteria are used in Criterion E to evaluate the product.

Evaluation criteria should follow SMART principles:

Specific	Clearly defines what is being tested — no vague language.
Measurable	Can be quantified or tested with a clear pass/fail outcome.
Achievable	Realistic, given the developer's skills and available time.
Relevant	Directly addresses the problem statement.
Time-bound	Can be tested within the project timeline.

EXAMPLE CRITERIA - ATTENDANCE SYSTEM

- "The system must correctly record attendance for all 30 students in a class within 60 seconds." **SMART**

- "The system must reject any date entered in a format other than DD/MM/YYYY and display the message 'Invalid date format'." **SM**
- "The attendance report must display correctly on a screen resolution of 1366×768 pixels." **SMA**

Not a SMART criterion: "The system should be user-friendly." This cannot be objectively tested. Rewrite as: "90% of test users must be able to record attendance for a class of 30 without consulting a help document."

Worked example: school library booking system

The following is a complete problem specification for a realistic scenario. Read each section and consider how it meets the requirements of that element.

Scenario: A school library has six study rooms. Students currently book rooms by writing their name on a paper sign-up sheet on the library door. Rooms are frequently double-booked. The librarian spends significant time resolving disputes and has no record of usage patterns.

Problem statement	The school library's paper-based room booking system results in frequent double-bookings and provides no usage data. The librarian spends an estimated two hours per week resolving booking disputes. Students lose study time when their reserved room is occupied by another group.
Constraints	TECHNICAL The solution must run on the school's existing computers (Windows 10, 4GB RAM). It must not require students to install software; it must run in a web browser. OPERATIONAL The solution must be ready for use at the start of the next academic term. The developer has no budget for paid hosting, so it must run on the school's local

server.

LEGAL

Student names and booking data must be stored in compliance with the school's data protection policy. Data must not be accessible to other students.

Objectives

To provide students with a reliable way to reserve a library study room in advance.

To eliminate double-bookings and reduce the time the librarian spends managing room allocation.

To give the librarian access to room usage data to inform future resource planning.

Inputs

Student ID (integer, 5 digits, entered via login form); booking date (DD/MM/YYYY, selected from a date picker); room number (integer, 1–6, selected from a list); start time and end time (HH: MM format, selected from a dropdown in 30-minute intervals).

Outputs

Booking confirmation displayed on screen, including room number, date, and time slot. Error message if the requested slot is already taken: "Room [n] is unavailable at that time — please choose another slot." Monthly usage report (on-screen table) accessible only to the librarian, showing bookings per room per week.

Evaluation criteria

1. The system must prevent any two students from booking the same room for an overlapping time on the same date. Testing: attempt 20 conflicting bookings - zero must succeed.
2. A booking must be confirmed or rejected within 2 seconds of submission when tested on the school's local server.
3. The system must reject any date entered outside the current academic year and display a clear error message.
4. The librarian's usage report must display the correct total number of bookings per room when tested against a dataset of 100 known bookings.

Summary

Problem statement

Constraints

Describes what the problem is and who is affected. No solution language.

Ask: What is the problem, and why does it matter?

Fixed conditions the solution must work within - technical, operational, legal.

Ask: What can the developer not change?

Objectives and goals

Broad aims of the solution - what it should achieve in general terms.

Ask: What is the purpose of this solution?

Input/output specifications

Precise definition of data entering and leaving the system, type, format, and range.

Ask: what goes in and what comes out?

Evaluation criteria

Specific, measurable, SMART standards are used to test whether the solution succeeds. Used in IA Criterion E.

Ask: How will I know if my solution works?

A common exam error is to write objectives that are actually evaluation criteria, or to write a problem statement that describes the solution. Keep these elements clearly separated; each answers a different question about the problem.

Flashcards

Use this Quizlet to learn all the key terms in this section.



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

Which of the following is NOT a key element of a problem specification?

A. ☐ Problem statement

B. ☐ Constraints

C. ☐ Goals

D. ☒ Algorithm design

2

A good problem statement should primarily focus on:

A. ☐ The tools and technologies that will be used.

B. ☐ The specific steps to solve the problem.

C. ☐ The desired outcome of the solution.

D. ☒ The clear and concise description of the issue.

3

Constraints in a problem specification refer to:

A. ☐ The goals that need to be achieved.

B. ☐ The different inputs that the solution will accept.

C. ☒ The limitations and restrictions that must be considered.

D. ☐ The expected outputs of the solution.

4

Why is it important to define input/output specifications?

A. ☐ To identify potential errors in the code.

B. ☐ To estimate the time required to solve the problem.

C. ☐ To determine the programming language to be used.

D. ☐ To clearly understand the data involved and the expected result.

5

Which of the following is an example of a constraint in the "Vending Machine" scenario?

A. ☐ The vending machine accepts only specific coins.

B. ☐ The vending machine should provide change.

C. ☐ The vending machine should dispense the correct snack.

D. ☐ The user should enter the snack code.

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